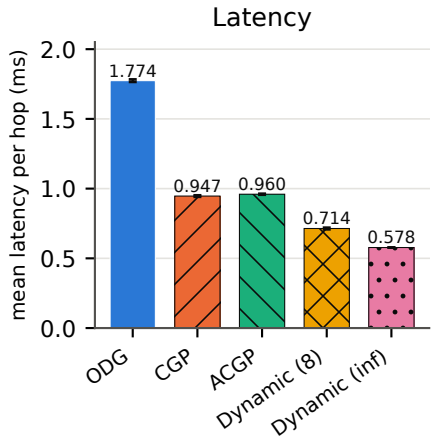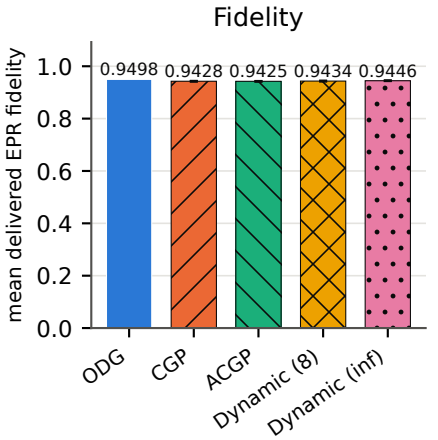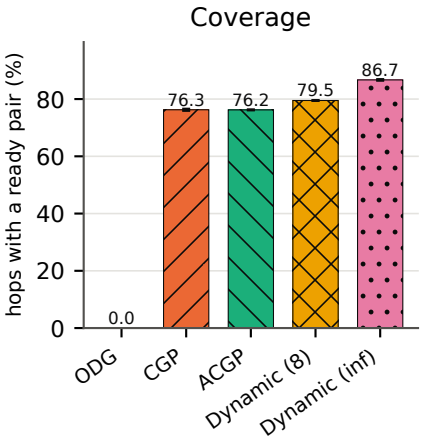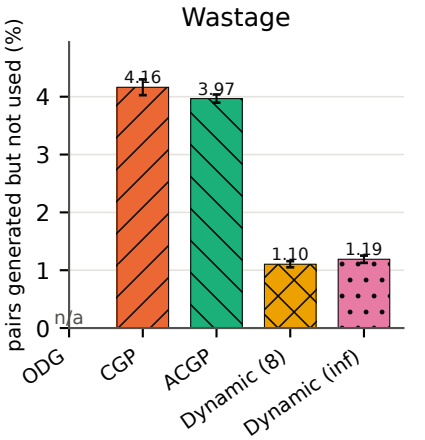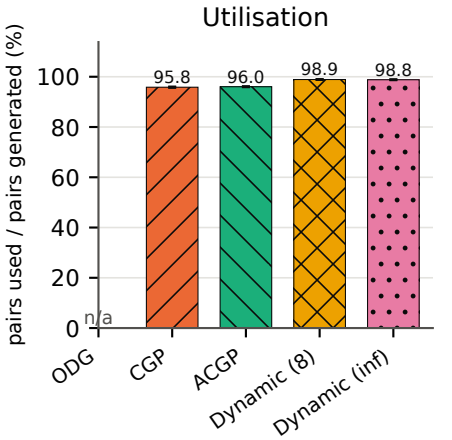QPE (exact), history routing: 8 memories per core, 6 for pre-generation